

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Reviewing the basics of ALU



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations.
Read and write about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can read about ALU.
- I can write sentences correctly about ALU.
- I can understand what ALU is in a written text.

Ask your partner:

- What kind of pieces make part of the CPU?
- What's the importance of a circuit in a CPU?





Let's Explore!

ALU, the Different World of Circuits



Instructions: Read the text below.

Please, welcome Mr. Kevin M.

Kevin M: One of the most important parts of the Central Processing Unit is the Arithmetic Logic Unit, let's call it ALU. ALU is a complex digital circuit that performs basic mathematical operations and calculations such as: addition, subtraction, multiplication, and division. And some other complex operations that require more space in CPU; when ALU is working it disperses heat and energy, now you know why computers get hot when working.

John Von Neumann proposed the ALU concept in 1945. Some of the most important functions of ALU are first, bitwise, that's the operation of individual bits of binary rules. Secondly, mathematical operations with binary numbers, that's the base 2 number system and the primary language for computers. ALU functions are necessary processes in order to work correctly, the CPU processes most of the data, so ALU is an important and vital part of data processing.

ALU can also perform logic operations with values NOT, AND, OR.

ALU data processing speed can be fast or slow, depending on the processor, its quality and price; you have this cheap and old processor that is very slow but you can retrieve your data processing speed if you have a good quality processor that transfers the data fast enough in order to make fast and successful transfers, that sounds expensive but if you want quality, that's what you have to do, buy an expensive but effective processor.

ALU is an important part of the computer system, without it, operations would become hard to solve.

Good maintenance is a key element if you want to keep your CPU clean, fast and powerful. Your computer is not an object, it's also your friend... Don't forget that! Thank you very much!



Let's Practice!

The ALU system functions.

Instructions: write the correct word to complete de sentence.

(That / These / This) are the most important functions of ALU: First, bitwise, (those're / these're / that's) the operation of individual bits of binary rules. Second are mathematical operations with binary numbers, the base of 2 number system, (that's / this's / it's) and the primary language for computers.



Let's Practice!

Talking about those operations the ALU can process.

Instructions: Write the words into the correct category.

What ALU can process	What ALU cannot process

1. NOT, AND, OR,
2. Quality
3. THIS, THAN, THOSE, THESE
4. Logic operations
5. WITH, SUCH, BUT
6. bitwise



Extension Activity

What ALU system is.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the ALU system functions.
- Design a post whit the origin, definition and functions of the ALU system.
- Share the poster with your teacher and classmates.

Self-Assessment

Yes No

- I look for more information about the ALU system functions.
- I design a post whit the origin, definition and functions of the ALU system.
- I share the poster with your teacher and classmates.

☐ ☐
☐ ☐
☐ ☐


Target Words

- **let's** (contraction)
we use this expression to invite someone to do something
e.g. One of the most important parts of the Central Processing Unit is the Arithmetic Logic Unit, let's call it ALU.
- **primary** (adjective)
primary means when something is important, when something goes first; rank, importance
e.g. binary numbers, that's the base 2 number system, and the primary language for computers.
- **CPU** (noun)
this is the brain of the computer, this is where the calculations occur
e.g. the CPU processes most of data, so ALU is an important and vital part of data processing.
- **enough** (adverb)
sufficient, when there is an amount of something you don't need anymore
e.g. a good quality processor that transfers the data fast enough.
- **successful** (adjective)
High levels of achivement
e.g. you have a good quality processor that transfers the data fast enough in order to make fast and successful transfers.
- **Maintenance** (noun)
preserve something
e.g. Good maintenance is a key element if you want to keep your CPU clean.



Learning Points

Can – Modal Verb

Can is a very important verb, it's an auxiliary verb and also a modal verb, we use *can* in several contexts and situations and it's a very helpful verb we can count on.

Can

We use *can* when we make a reference about **abilities**, we use it to say if we are able to perform an action or not, depending on the possibilities and depending on the circumstances.

We use *can* when we **give permission** or when we **ask for permission, the verb *can*** is less formal and more common to use it with friends or an acquaintance for example.

We use *can* when we make a **deduction**, when we know something in advance and then we are able to figure out what possibly happens next

Structure

Subject + Can + Verb in present (Infinitive) + Complement

NOTE:

In negative form we just change **Can** for **Can't**

In interrogative form we switch **Subject** and **Can**, *can* goes first and then next to it is the subject.

Examples

- *ALU can also perform logic operations with values NOT, AND, OR*
- *ALU data processing speed can be fast or slow*



Learning Points

Demonstrative Pronouns

We use demonstrative pronouns when we make a reference about two things specifically:

1. Singular or Plural
2. Distance

Demonstrative pronouns

(This, That, These, Those)

When we make a singular sentence with (1 person, 1 object) we use demonstrative pronouns **this** or **that**

This = **singular** form and when a person or an object is **near**

That = **singular** form and when a person or an object is **far**

When we make a plural sentence with (+2 persons, +2 objects) we use demonstrative pronouns **these** or **those**

These = when **plural** and when persons or objects are **near**

Those = when **plural** and when persons or objects are **far**

Examples

- *Now you know why computers get hot when working, that's because of ALU.*
- *That's what you have to do, buy an expensive but effective processor.*